

Data Sommmunication

1st Class

1. Define signal

Answer: 3 no slide 1st para with diagram

2. How signal is transform in frequency?

Answer: 4 no slide with diagram

3. Compare discrete time signal and digital signal

Answer: 8 no slide, 2nd para with diagram and slide 9 full

4. Draw the complete diagram of Analog to Digital Conversion

Answer: Slide 13.

5. State Nyquist sampling criteria?

Answer: Slide 17.

Nyquist Sampling Criteria (সহজ ভাষায়)

কথাটা কী?

কোনো continuous (একটানা) সিগন্যালকে যদি আমরা ডিজিটালভাবে নিতে চাই (sampling করি), তাহলে **sampling frequency** এমন হতে হবে যেন আসল সিগন্যালটা আবার ঠিকভাবে ফেরত পাওয়া যায়।

If we want to take a continuous signal digitally (i.e., sample it), then the sampling frequency must be such that the original signal can be perfectly reconstructed again.

6. Draw the Spectrum

Answer: Example slide no 18

2nd Class

7. How quantization is done and what its impact on received signal

Answer: Slide 19 full with diagram.

8. Show PCM technique with diagram.

Answer: 20 no slide with diagram.

9. Give the concept of DPCM and Delta modulation with diagram?

Answer: Slide 25 and 26 with diagram.

10. Problem on line coding.

Answer: Slide 31 but polar, unipolar, Bi polar egulor definition asbe na just bujhar jonno dewa ache.

11. Define multiplexing and multiple access?

Answer: Slide 41 with diagram.

12. Give the concept of Frequency division multiplexing(FDM).

Answer: Slide 43 1st diagram with 1st para.

13. Give the concept of Time division multiplexing (TDM).

Answer: 1st diagram with 1st para with yellow color.

Computer Network

14. What is the function of modem in Computer Network?

Answer: 1st Slide, 1st para.

15. Why Digital modulation is necessary in wired communication?

Answer: 2nd slide, para 1,2

16. Why Digital modulation is necessary in wireless communication?

Answer: 3rd slide with any 3 para of all.

3rd Class (2. Lec 2 625P.pdf)

1. Write basic principle of BFSK.

Ans: Slide 4 er uporer shadow color not block diagram

2. Draw BFSK modulator circuit.

Ans: Slide 5 Only figure 1 diagram not description.

3. Draw Coherent demodulator circuit of BFSK.

Ans: Slide 6 Only diagram and with $\cos(2\pi f_1t) - \cos(2\pi f_2t)$

4. Give basic principle of BPSK.

Ans: Slide 8 purotai

5. Why M-ary PSK is preferable compared to BPSK?

Ans: Slide 11 er uporer shadow color only

6. Derive the initial phase angles of 8-PSK.

Ans: Slide 11 er nicher 2 line only. In 8 PSK evabe ans start korbo.

7. **Why Quadrature amplitude modulation is widely used in data communication? Give the expression of modulated wave?**

Ans: Slide 18 er shadow lekha and first equation only.

8. **Draw a simplified waves/waveshaps**

Ans: Slide 19 diagram.

4rth Class (Lec 3 625P.pdf)

1. **Draw three layer architecture of two philosopher communication**

Ans: Slide 18 er . (Only figure)

2. **Mention two basic functions of Physical Layer.**

Ans: Slide 28 er 2nd and 3rd point

3. **Mention three basic functions of LLC (Logical Link Control).**

Ans: Slide 30 er nicher 3rd point only and 31 no slide er 2nd para and 32 slide er 1st para.

4. **Compare error detection and error correction.**

Ans: Slide 31 er 2nd and 3rd para.

5. **Compare physical and logical address.**

Ans: Slide 36 er 1st 2 para.`

5th Class(Lec 4 625P.pdf)

1. **What is the function of headers and trailers of a frame?**

Ans: Slide 3 er 1st para.

2. **Define frame with diagram.**

Slide 2 er diagram er uporer ongsho and diagram.

3. **Show bit stuffing with appropriate diagram.**

Ans: Slide 10 er diagram only.

4. **Compare bit and byte stuffing.**

Ans: 11 slide all.

5. **Give the idea of error correction.**

OR

Give the idea of ‘error correction through retransmission’ and

‘autonomous error correction’,

Ans: Slide 13 er 2nd and 3rd para

6. Write the graph of CRC algorithm.

OR

Write the steps of the Cyclic Redundancy Check (CRC) algorithm.

Ans: Slide 17 er green color.

7. Example-1

Ans: Slide 18 and 19.

8. How to avoid single bit error, two bit error and odd number of bit error in CRC?

Ans: Slide 24 and 25 1,2,3

9. What will be the probability of $e_x = g_x$ OR what will happen $e_x = g_x$

OR

What is the probability of CRC failing to detect error?

Ans: Slide 26 er 5 no point.

Class Test-2 Class 3 to Class 5 **porjonto**

6th Class (Lec-4)

1. Define flow control of LLC sublayer.

Ans: Slide 28 er 3rd para.

2. Give basic working principle of Stop and Wait flow control.

Ans: Slide 29

3. Draw few steps of Sliding Window flow control.

Ans: Slide 36,37 and 38 only diagram no description.

4. What is the main problem of random access protocol?

Ans: Slide 46 er seser parar “The efficiency theke of collision porjonto”

5. What is the main drawback of controlled access protocol?

Ans: 47 Slide 2nd para

6. Compare exhaustive service and gated service of token ring protocol.

Ans: 49 slide 2nd para and slide 50

7. Why CSMA/CA is used in wireless network?

Ans: 53 slide 1st para.

Class 7th and 8th
Class 8th Exam chilo tai only class 7th er question gulo likhe
dicilo. (Lec-5)

1. Draw a network (4 no slide er yellow color question).

Ans: 4 no slide er puro diagram only.

2. Mention the function differentiated service of IP packet.

Ans: Slide no 7 decription only.

3. Explain operation of IP fragmentation and reassembly with diagram.

Ans: Slide 10 & 11 er table bade sob.

4. Give the concept of tunneling with appropriate diagram considering connection of IPV6 routes through IPV4 routes.

Ans: 23 slide diagram only.

5. What is the subnet mask? What is it function.

Ans: Slide 26 er last para bade sob.

6. Problems on slide number 34 or Example of 34 slide.

Class 9th (Lec-6)

1. Compare datagram and virtual circuit.

→ Ans: Slide 3

2. Routing table of connectionless service

→ Ans: Slide 6 → (diagram dewa thakbe table banate hobe)

3. Routing table of connection-oriented service

→ Ans: Slide 9 → (diagram dewa thakbe table banate hobe)

4. Write 3 differences between datagram and virtual circuit

→ Slide 11

5. Compare non-adapting and adaptive algorithm

→ Slide 13 (last 2 para)

6. Define fairness and optimality of a network

→ Ans: Slide 14 (2nd and 3rd para)

7. **How they conflict in practical life.**
→ Ans: Slide 15 → diagram and description
8. **Problem on sink tree**
→ Ans: Slide 18, 19, 20
9. **Which parameters are considered to determine path length**
→ Slide 21 (6 points).
10. **Mention remedial measures of flooding in a network.**
→ Slide 33 (last 3 para)

Class 10th (Lec-7)

11. **Give the basic concept of logical & physical communication.**
→ Slide-3 (Only text)
12. **Draw three-way handshaking with appropriate diagram**
→ Slide-11 (Only diagram)
13. **How old duplicate CR is avoided in transport layer**
→ Slide-12 (Only diagram)
14. **How both old duplicate CR & old duplicate data are avoided in transport layer**
→ Slide-13 (Only diagram)
15. **Compare asymmetric & symmetric release**
→ Slide 14 2nd para.
16. **Draw three-way handshaking for connection release**
→ Slide-18 (Only diagram)
17. **Give the basic concept of multiplexing & demultiplexing at transport layer**
→ Slide-23 (Full – diagram & text)
18. **Give the idea of congestion window & receive window**
→ Slide 28 er first color portion bade sob.
19. **Draw the profile of TCP Tahoe**
→ Slide-31 (Only diagram)
20. **Draw the profile of TCP Reno**
→ Slide-34 (Only diagram)